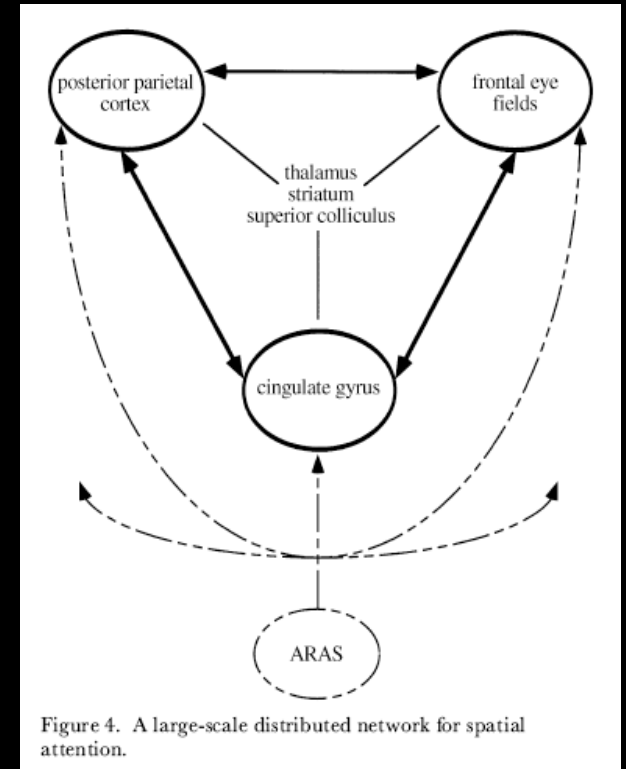
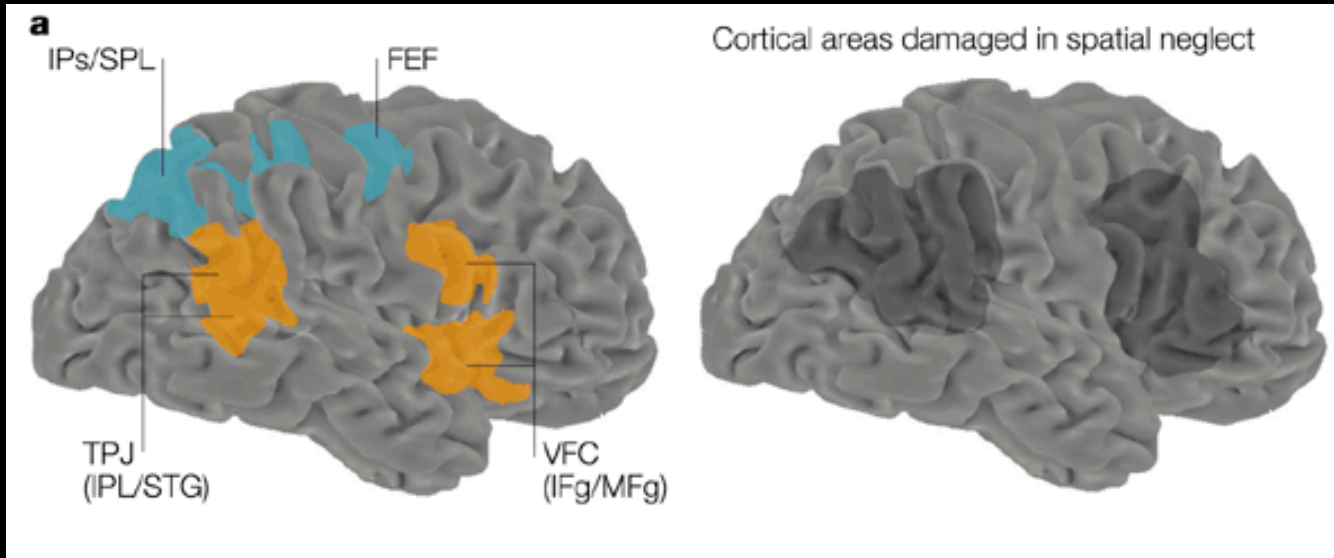


**Emotion and awareness for attentional factors in
neuropsychological patients**

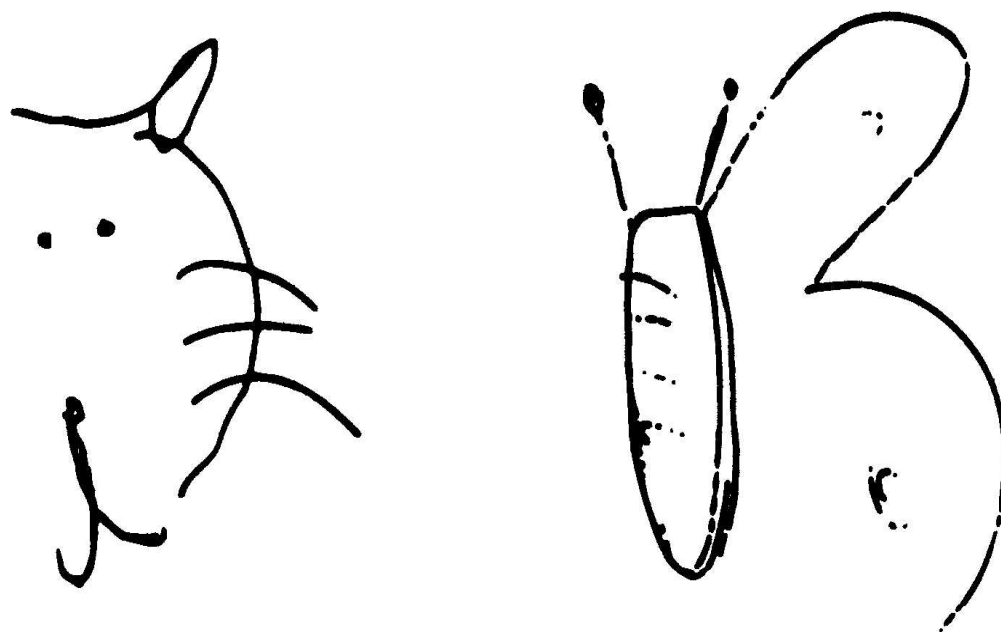
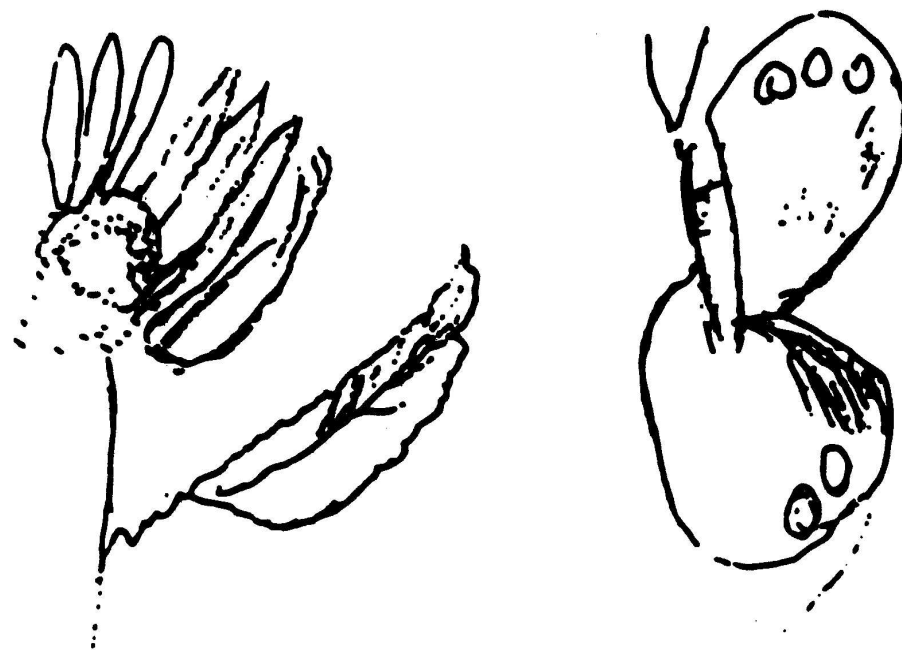
Anatomo-functional correlates of the attentional system in humans



Patients with Unilateral Spatial Neglect



Memory drawing



Federico Fellini

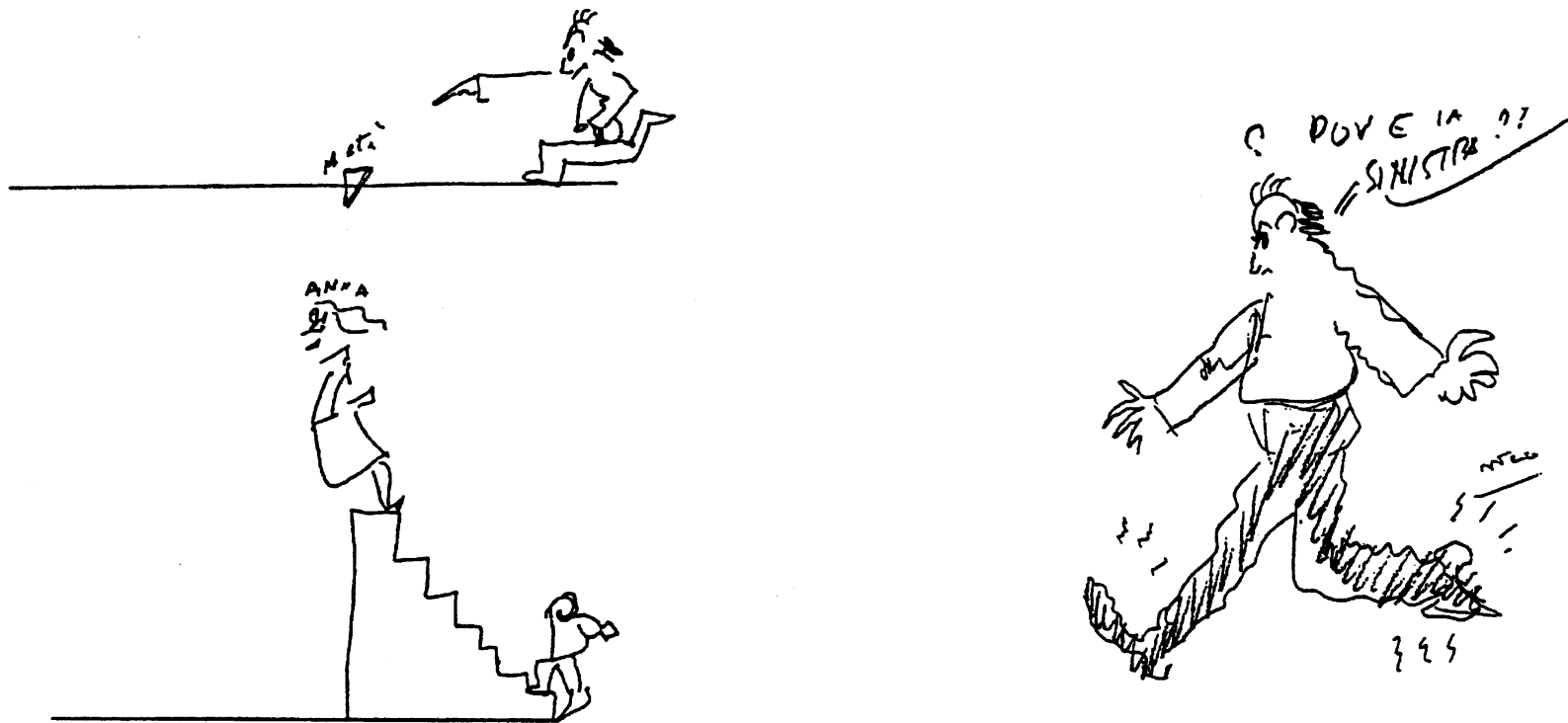
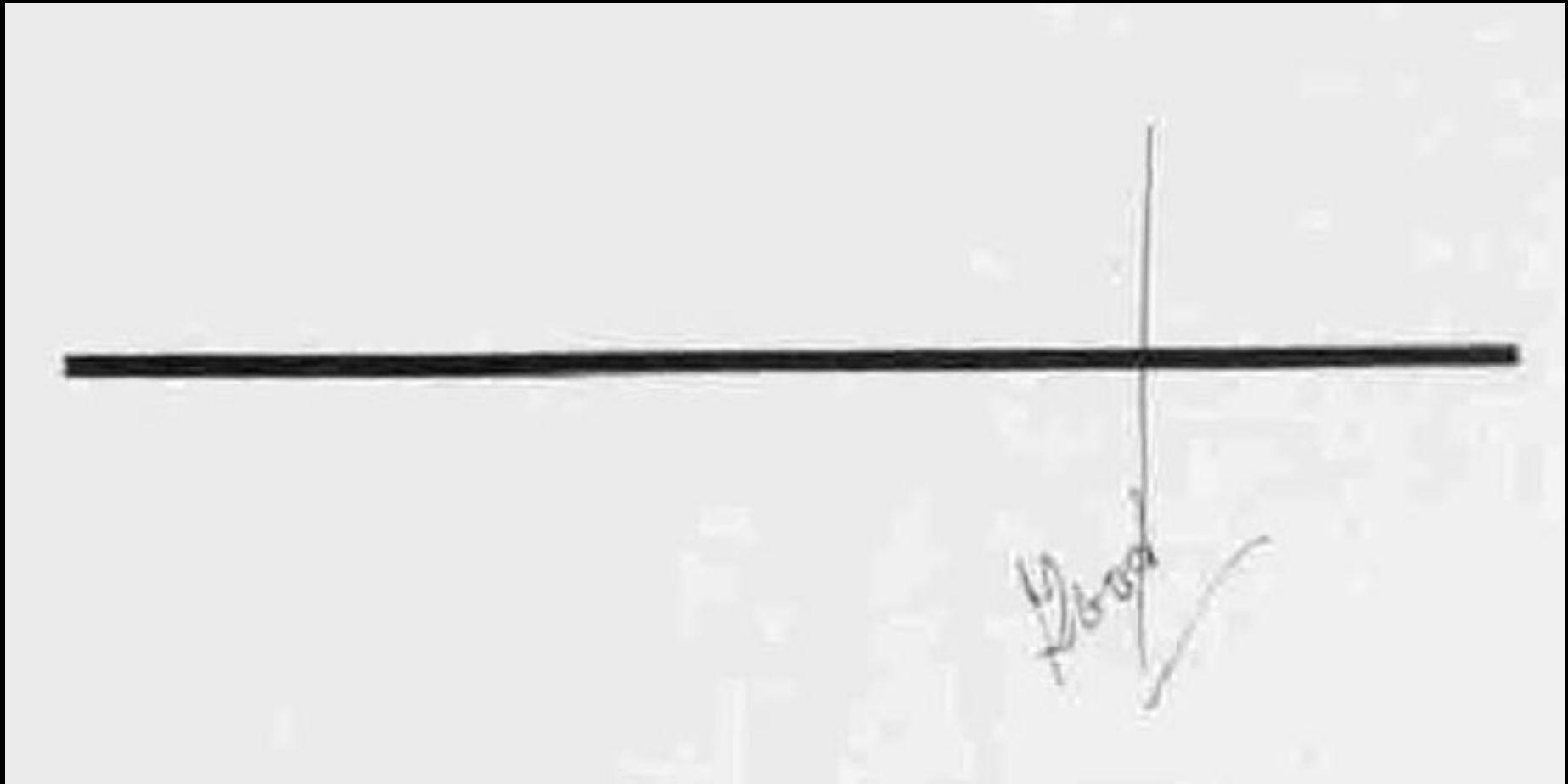
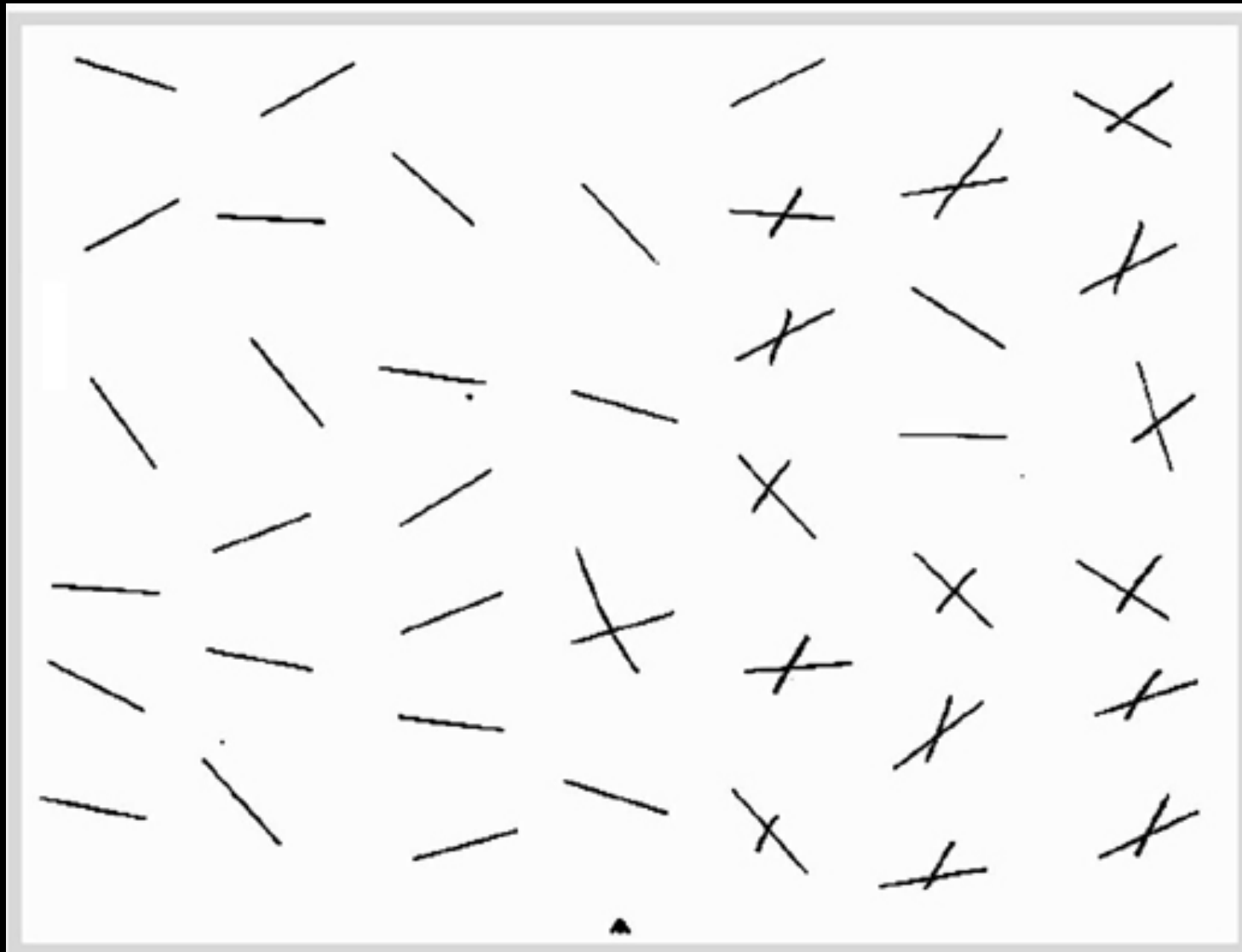


FIGURE 3–8. Left: Fellini's line bisection performance. The bisection point is rightward of true center. The whimsical embellishments are on the right side of both lines. The left sides of the lines remain unadorned. Right: Fellini looking for the lost left side of the world as he asks himself, "*Dov'è la sinistra?*" (Where is the left?) From Cantagallo and Della Sala.⁴⁵

Line bisection



Cancellation test





Anton Raederscheidt



Marshall & Halligan (1998)
Nature

Visual search and emotions in neglect

(Lucas & Vuilleumier (2008) Neuropsychologia)

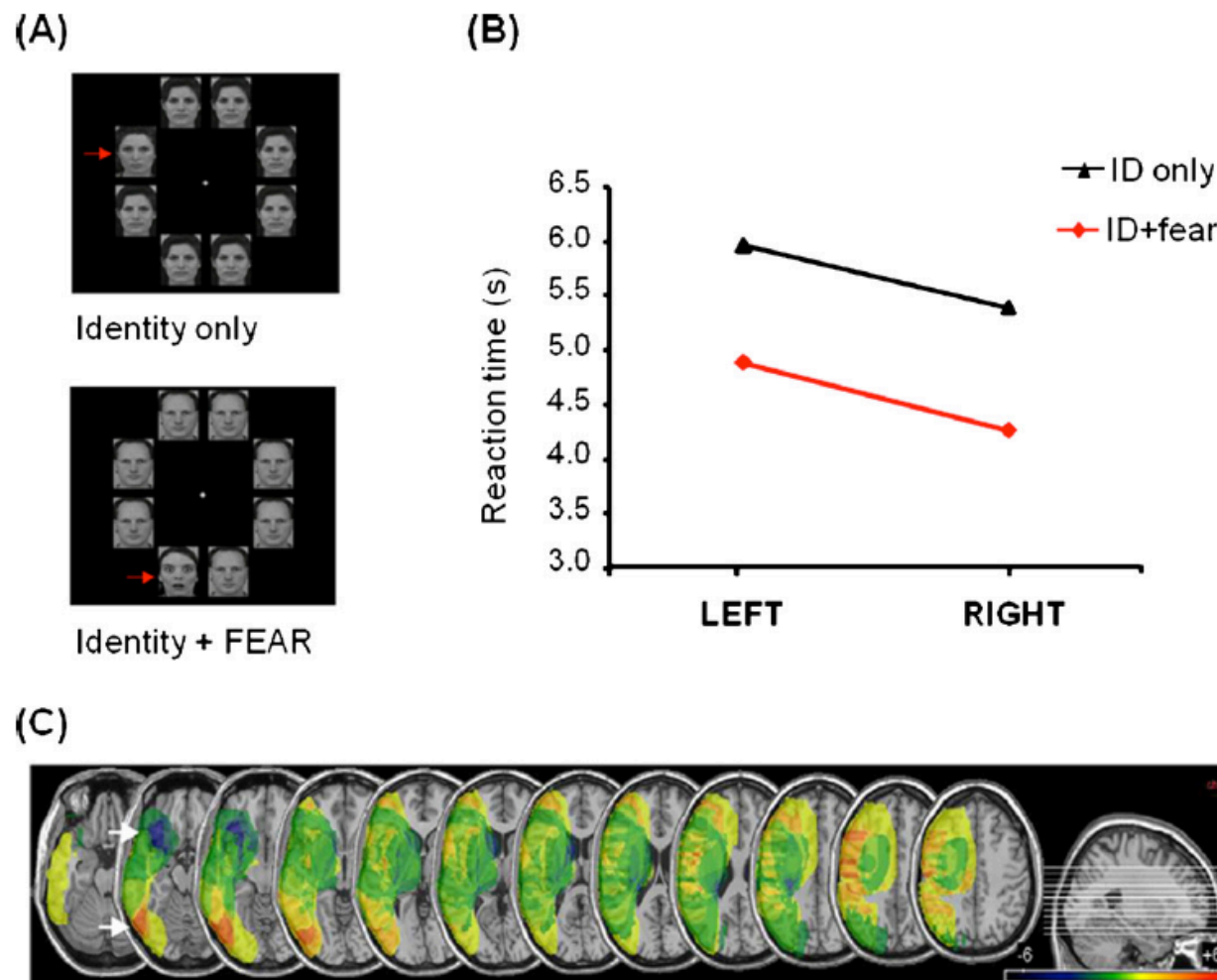
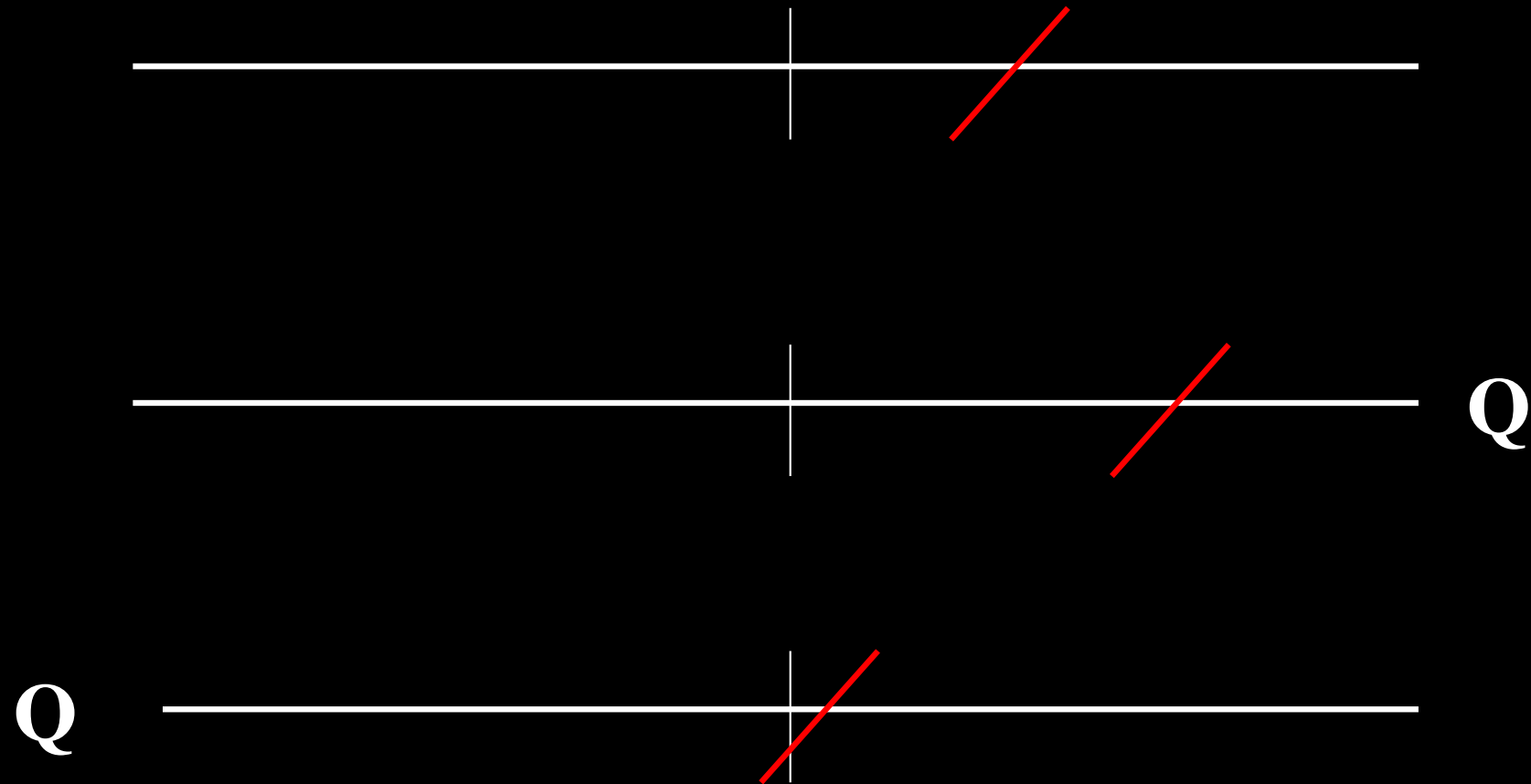


Fig. 2. Effects of emotion on visual search. (A) Illustration of two different conditions with either a neutral (upper) or fearful (lower) target in a task where neglect patients had to search and report the gender of the face with odd-out identity. (B) Average response times in 13 neglect patients showed better detection of face singletons, among 7 other neutral faces when those singletons depicted a fearful expression (therefore differing in both identity and expression; ID + FEAR), as compared to a neutral expression (that is, differing only in identity; ID only). This emotional cueing benefit was found for stimuli presented in both visual hemifields, in combination with generally slower response times to contralesional targets. Healthy controls showed similar emotional cueing effects, but no spatial asymmetry or even a left-side advantage (data not shown) (C) Lesion-symptom mapping in the same patients showing areas correlating with the magnitude of the difference in RTs for neutral minus emotional faces. Damage to the posterior temporo-parietal junction was associated with larger effects of emotional cues (red-orange), whereas anterior lesions in orbitofrontal regions was associated with smaller effects of emotional cues (blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

Adapted from Lucas and Vuilleumier (2008).

Bisections with cues

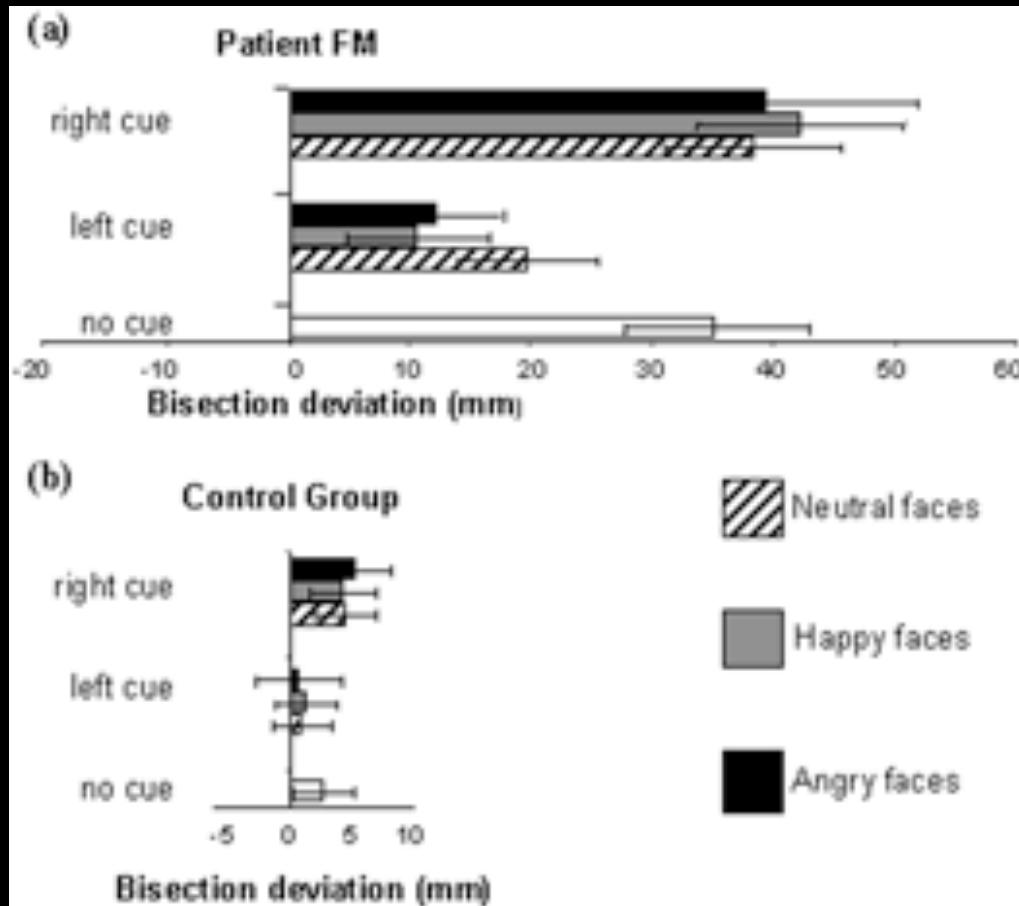
Neglect patients



Bisection task (prevision)



Emotional cues effect



Tamietto et al., 2005
Neurocase

Typical performance of a patient with visual extinction

One on
the right
side

One on
the left
side

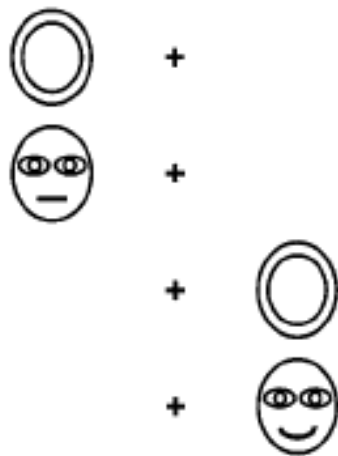
One on
the **right**
side



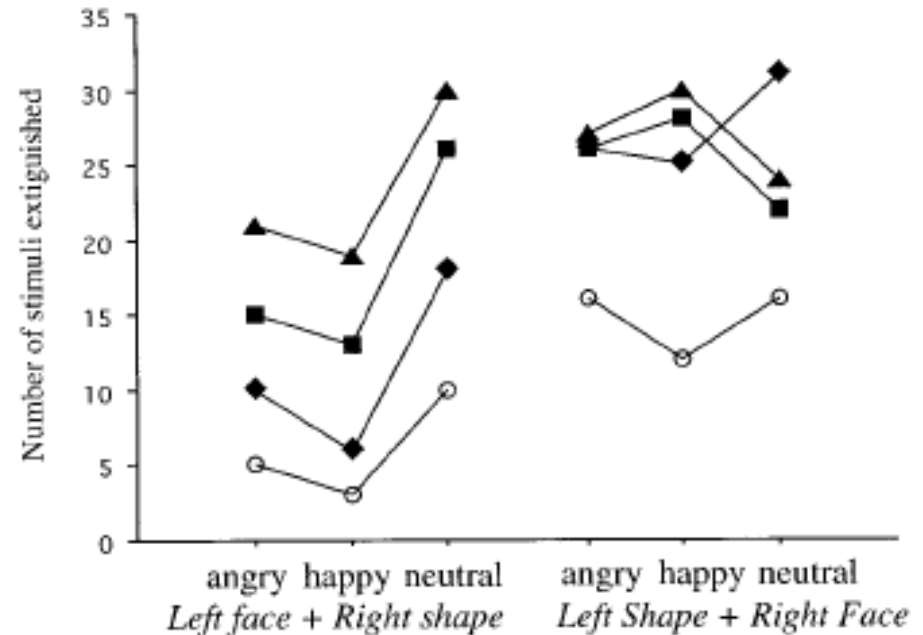
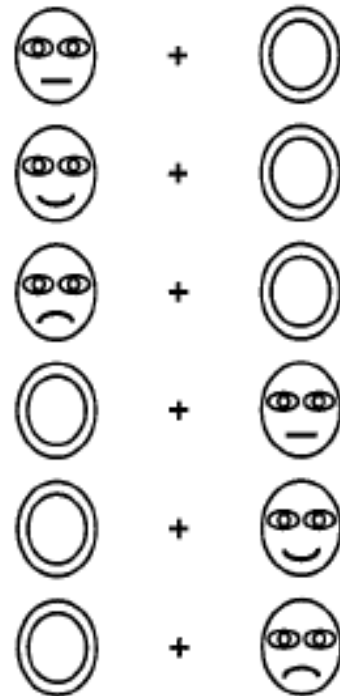
Emotions in a visual extinction paradigm

A. Experiment 1.

Unilateral trials



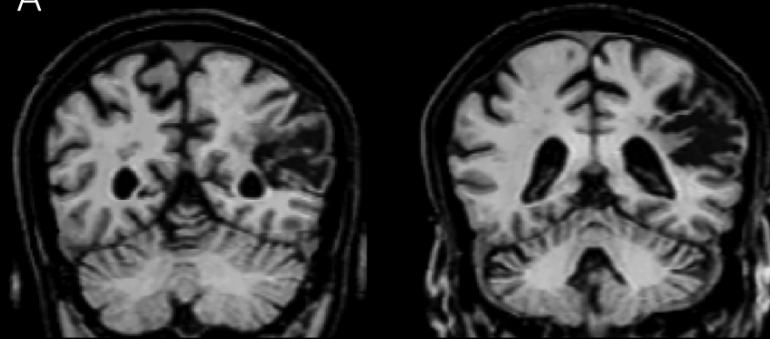
Bilateral trials



Vuilleumier 2001
Neurology

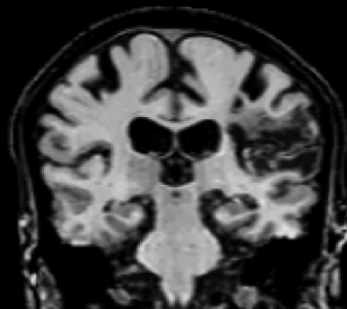
Patient MP

A

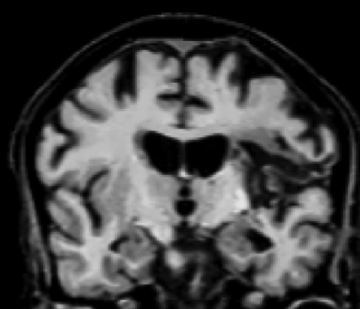


$Y = -55$

$Y = -40$

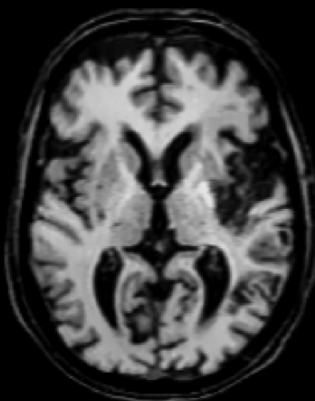


$Y = -25$

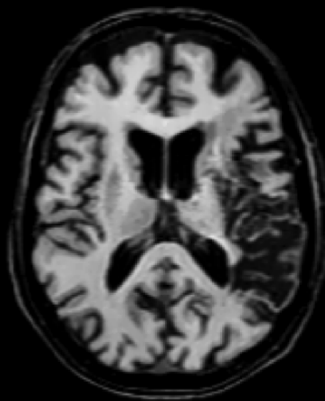


$Y = -10$

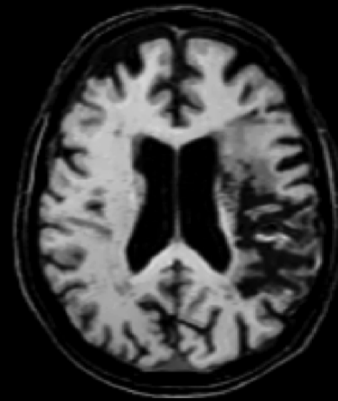
B



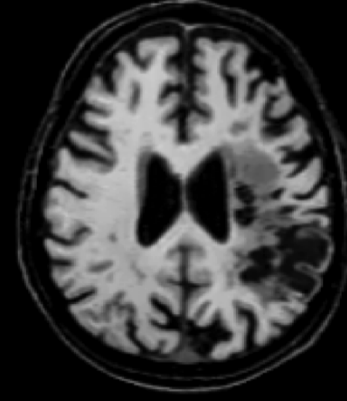
$Z = 6$



$Z = 14$

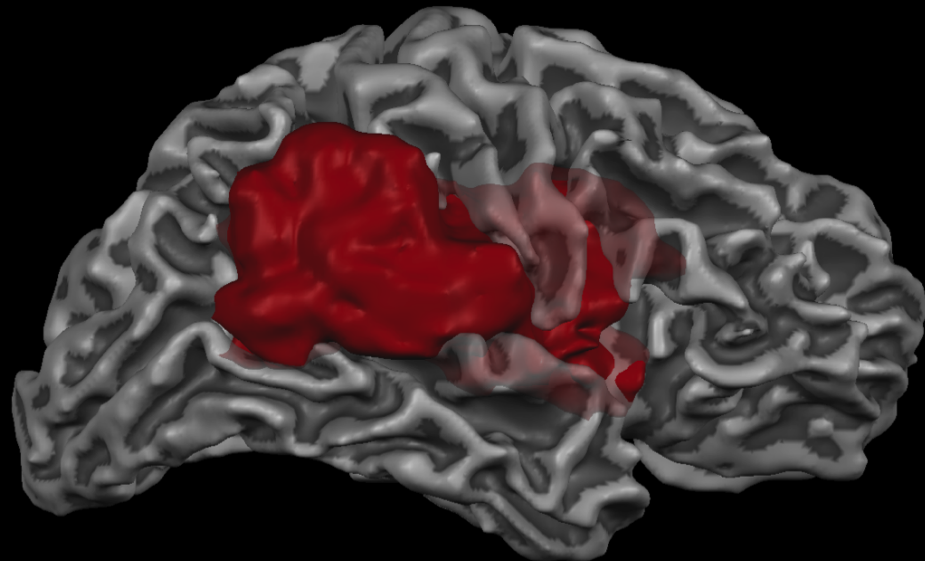


$Z = 22$



$Z = 30$

C



Attentional Unawareness for emotions in neglect

. “Emotional attention” alters perceptual sensitivity or response bias?

Behavioral and psychophysical evidence from signal detection theory

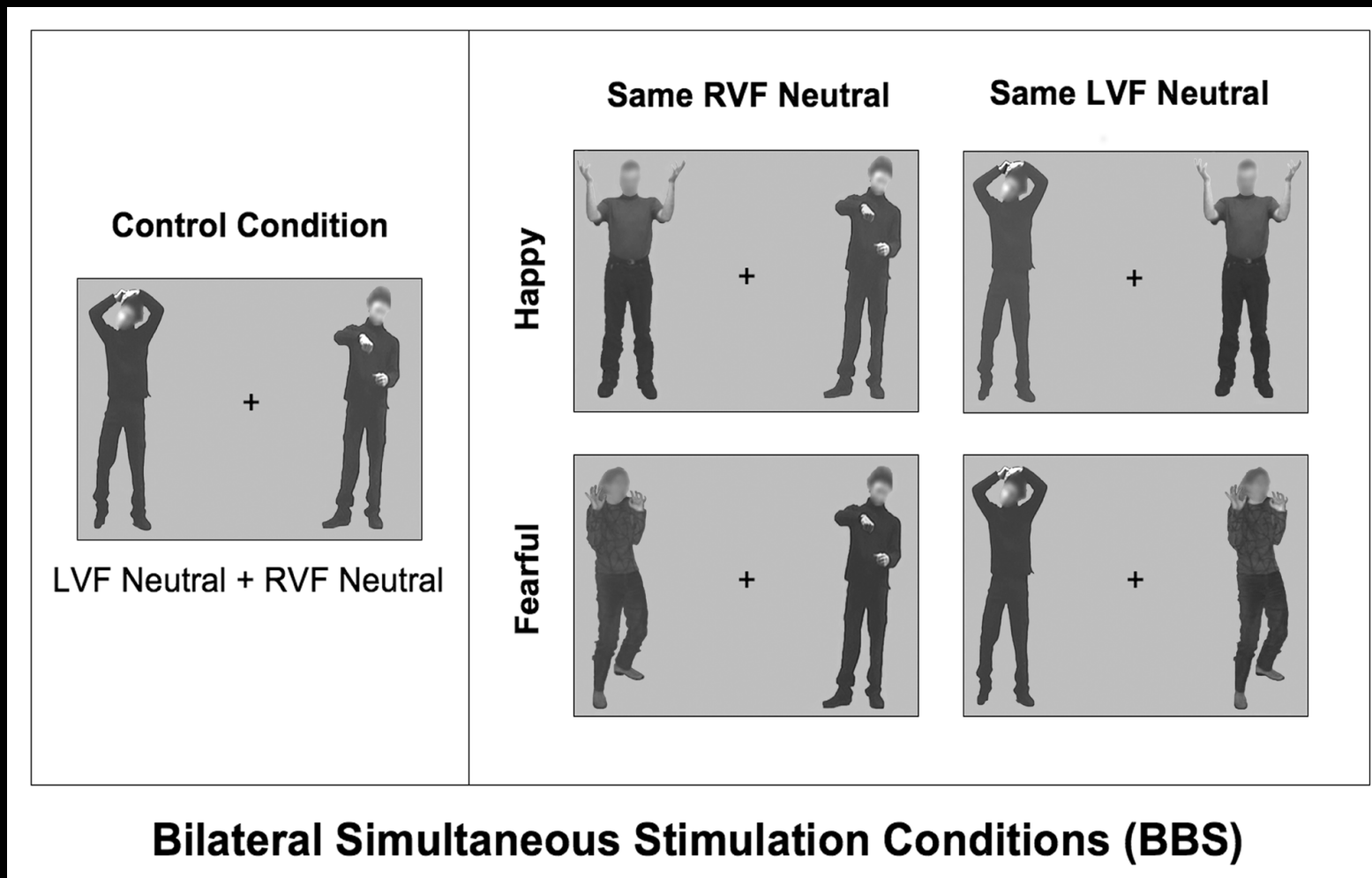
What is the relation between physiologic arousal and visual awareness for emotions?

psychophysiological evidence from pupil dilation data

What is the neural fate of unattended emotions and the neural fingerprint of attentional awareness for emotions?

fMRI evidence

Design



Task:

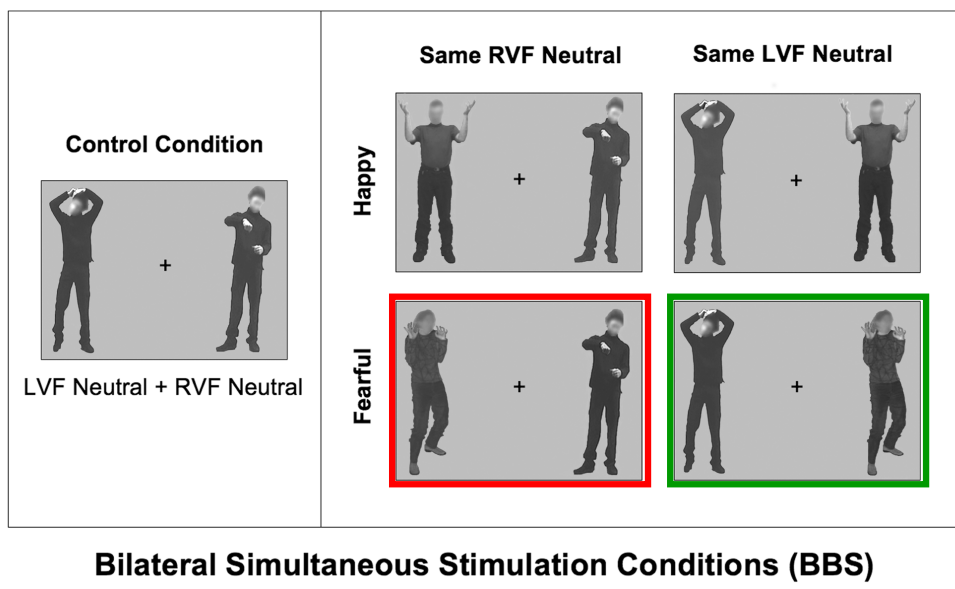
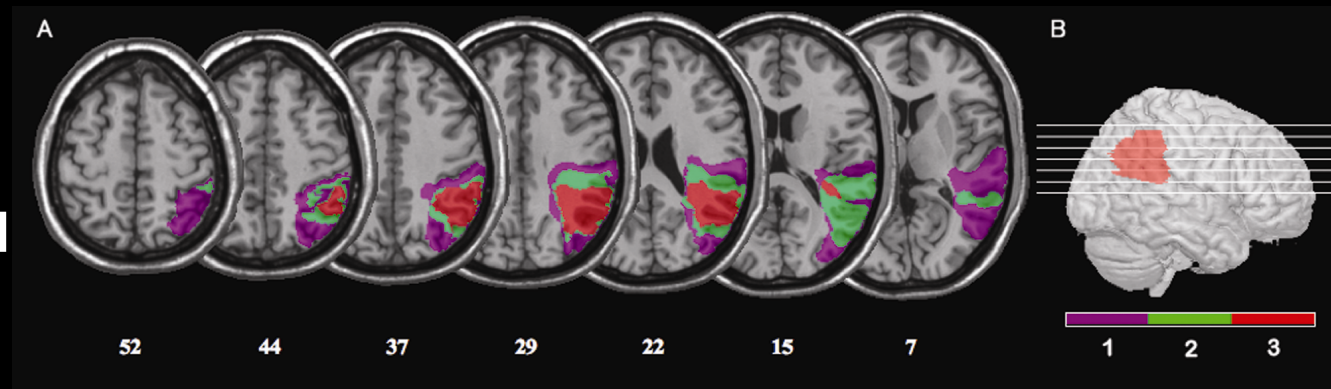
1. Report the number and location of stimuli

2. Rate confidence on 4 point scale from “least” to “most”

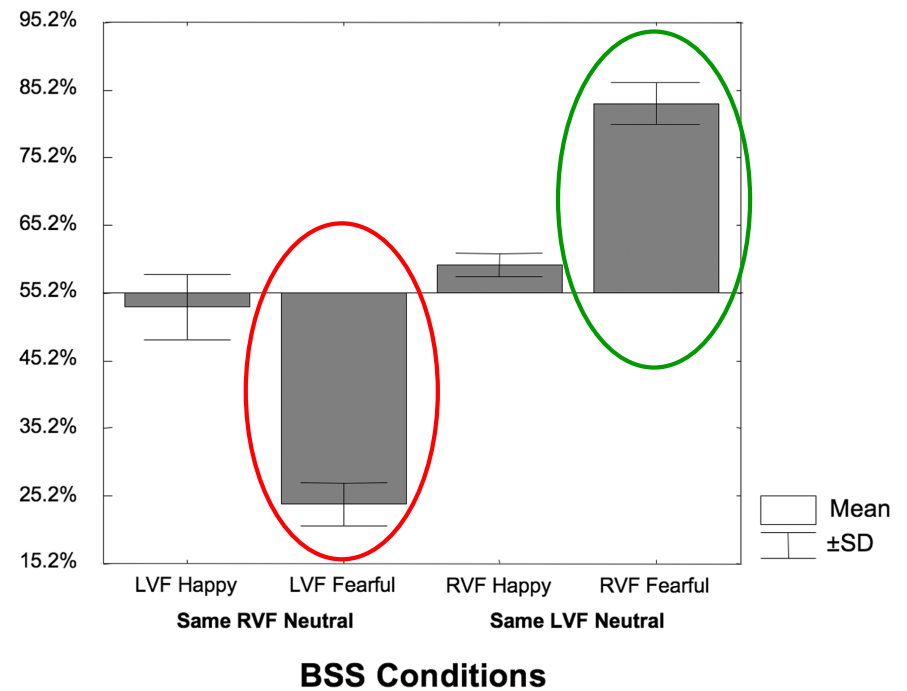
1	2	3	4	5	6	7	8
NO				YES			
(signal absent)				(signal present)			

3 Neglect patients:

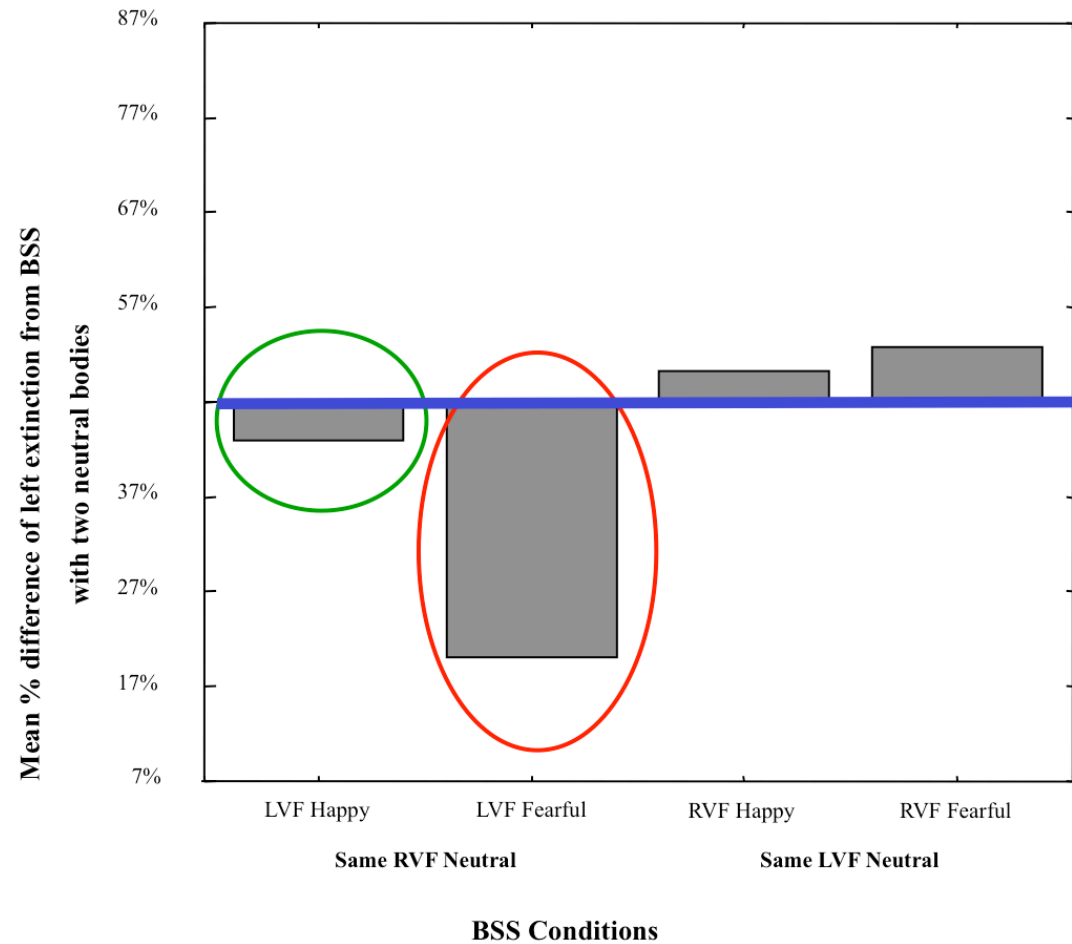
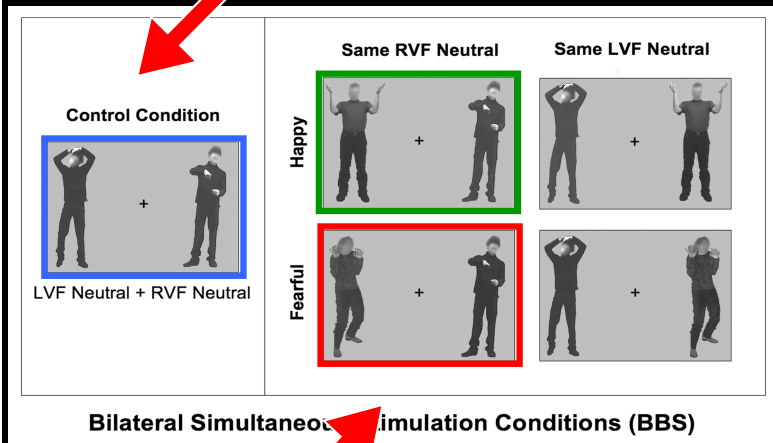
Pathological leftward inattention



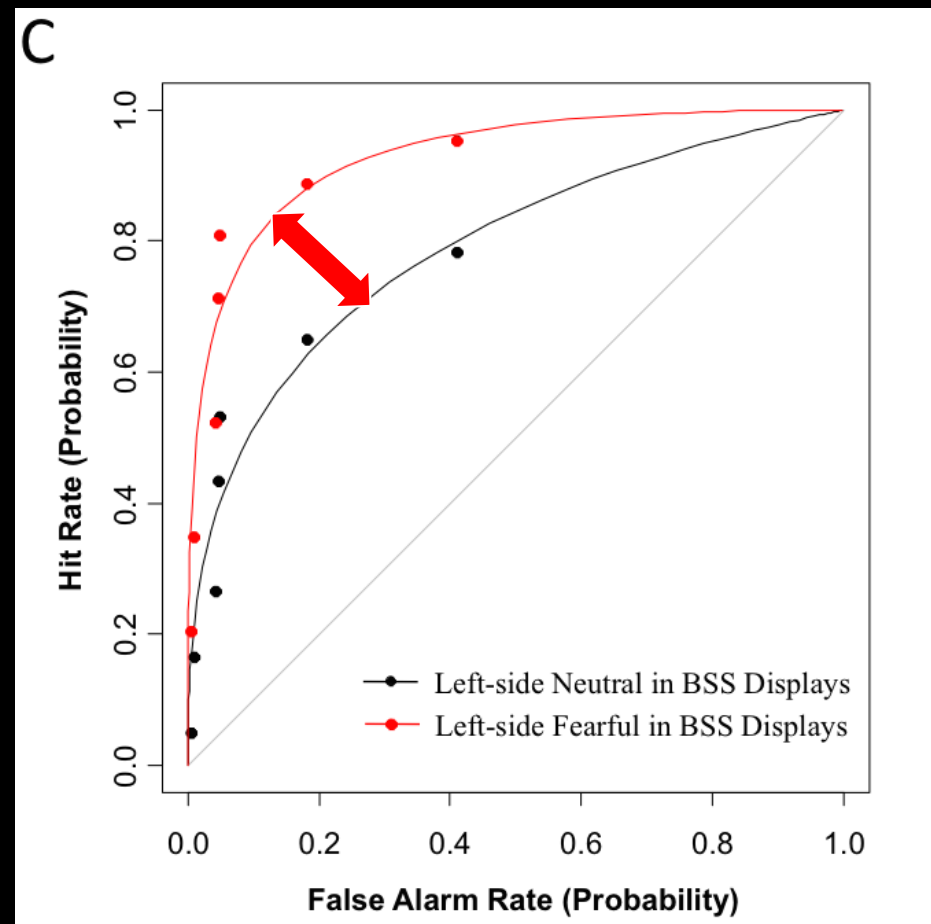
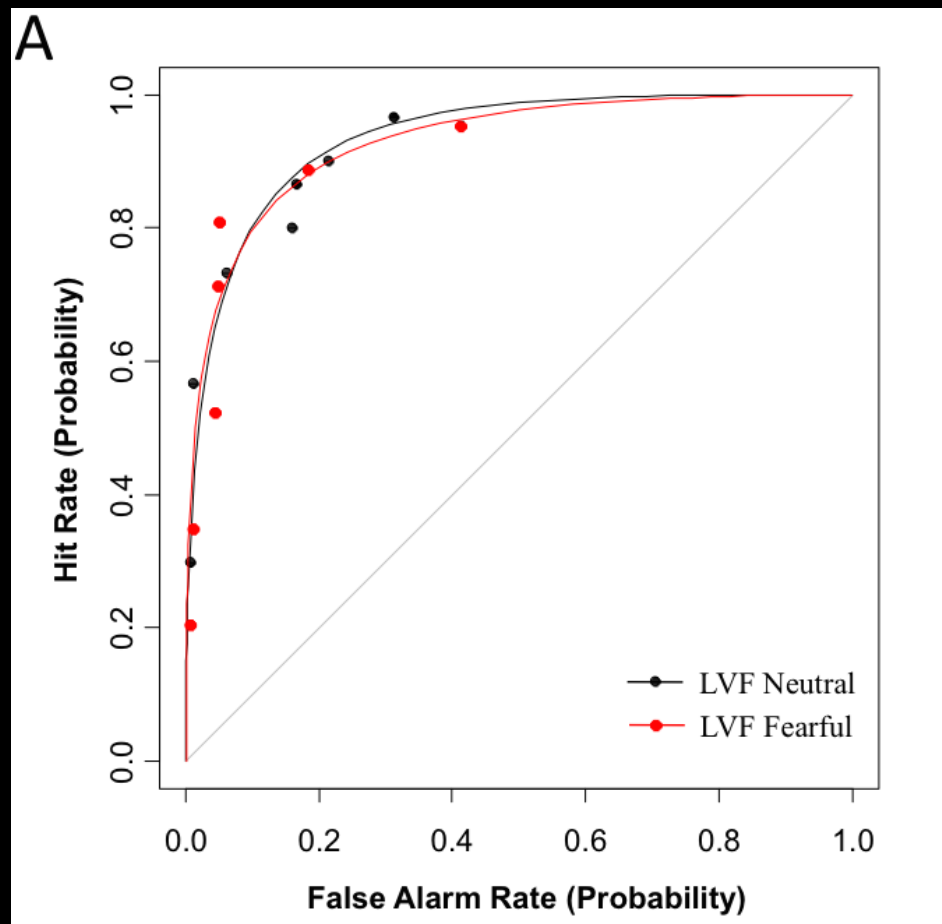
Mean % difference of left extinction from BSS with two neutral bodies



Behavioral Results



Signal Detection Analysis of ROC curves



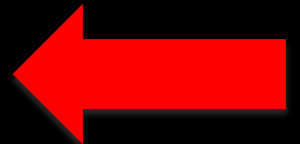
Attentional Unawareness for emotions in neglect

Emotional attention” alters perceptual sensitivity or response bias?

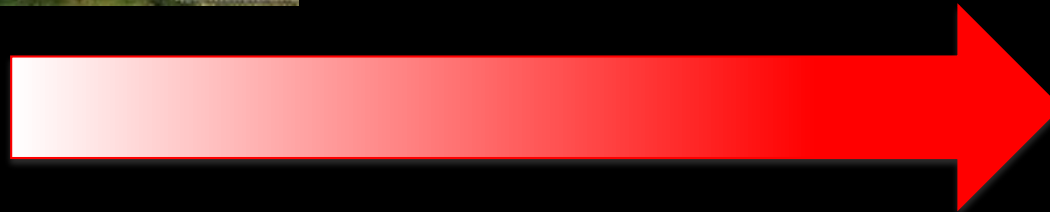
Emotional stimuli bias attention and promote visual awareness by enhancing perceptual sensitivity

What is the relation between physiologic arousal and visual awareness for emotions?

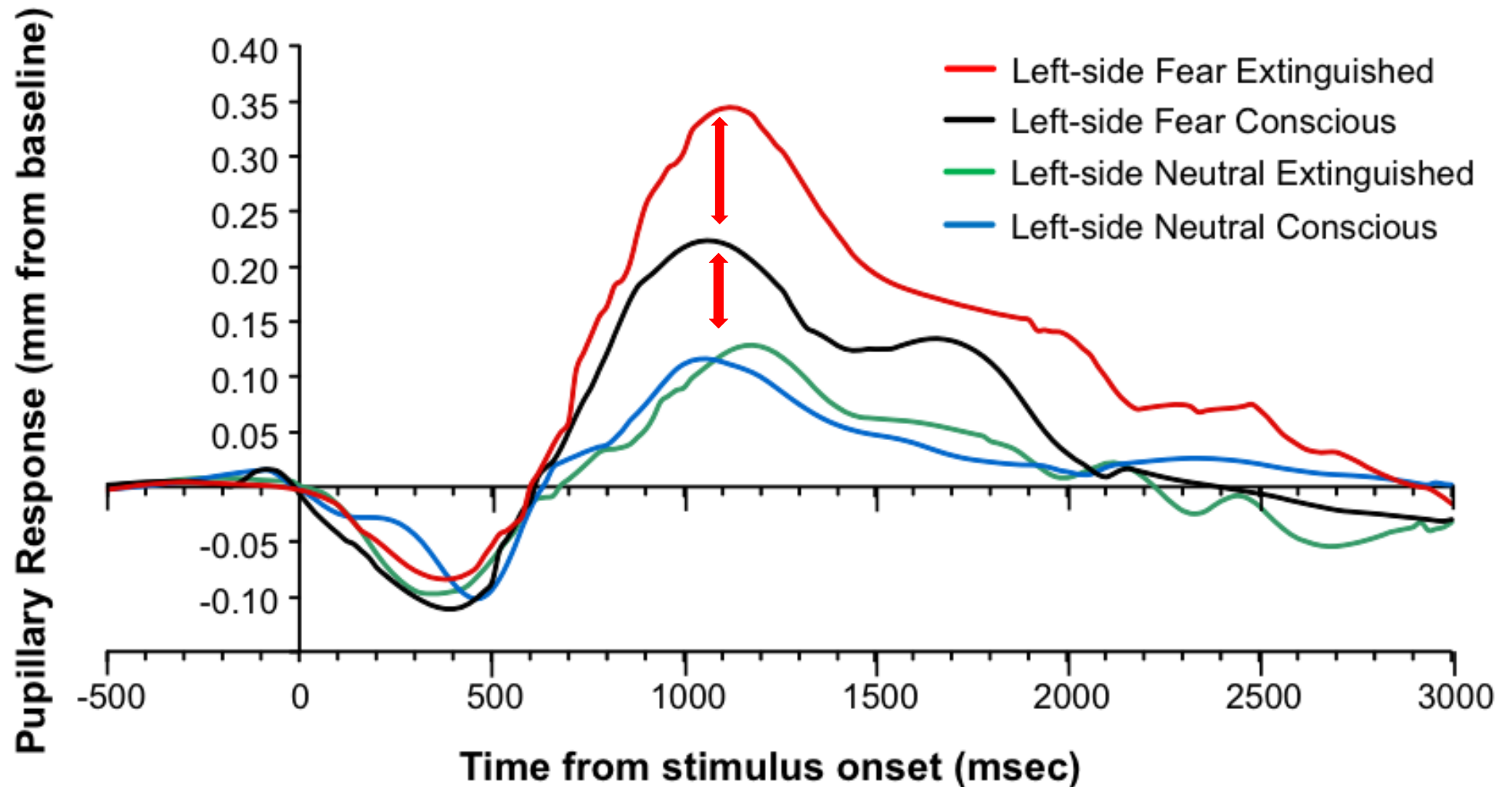
psychophysiological evidence from pupil dilation



Pupil dilation



Psychophysiological Results



Attentional Unawareness for emotions in neglect

Emotional attention” alters perceptual sensitivity or response bias?

Emotional stimuli bias attention and promote visual awareness by enhancing perceptual sensitivity

is the relation between physiologic arousal and visual awareness for emotions?

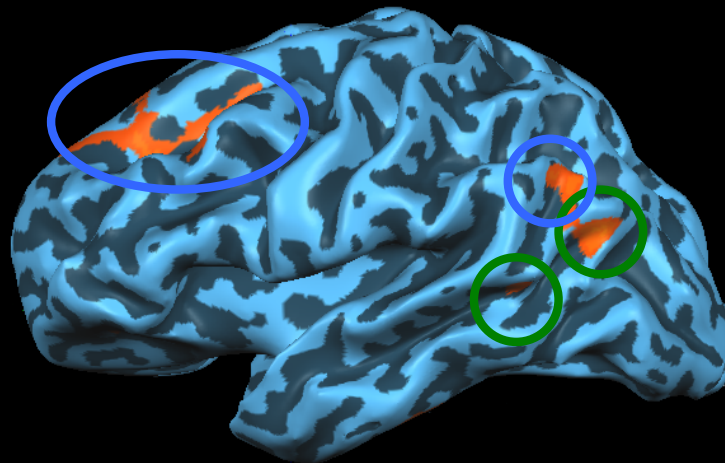
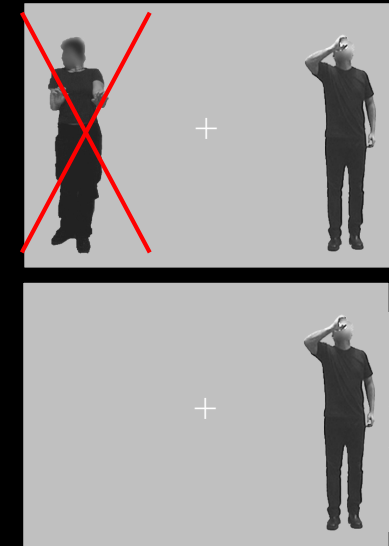
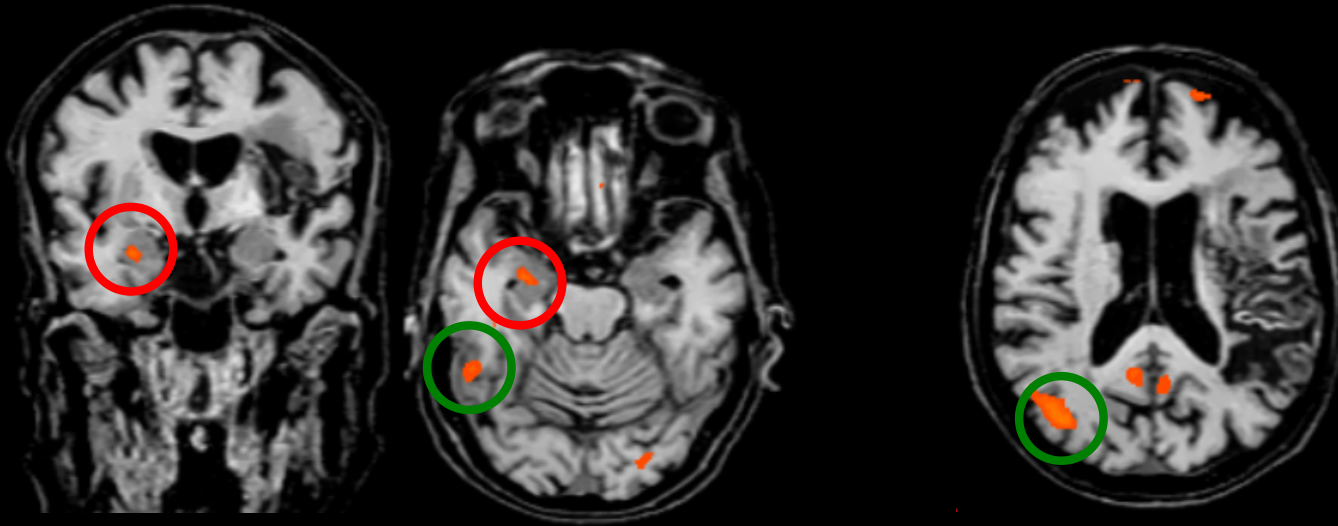
Additive effects of emotion and awareness on peripheral changes. Decoupling between (increased) somatic changes and (lack of) visual awareness for emotions

What is the neural fate of unattended emotions and the neural fingerprint of attentional awareness for emotions

fMRI evidence



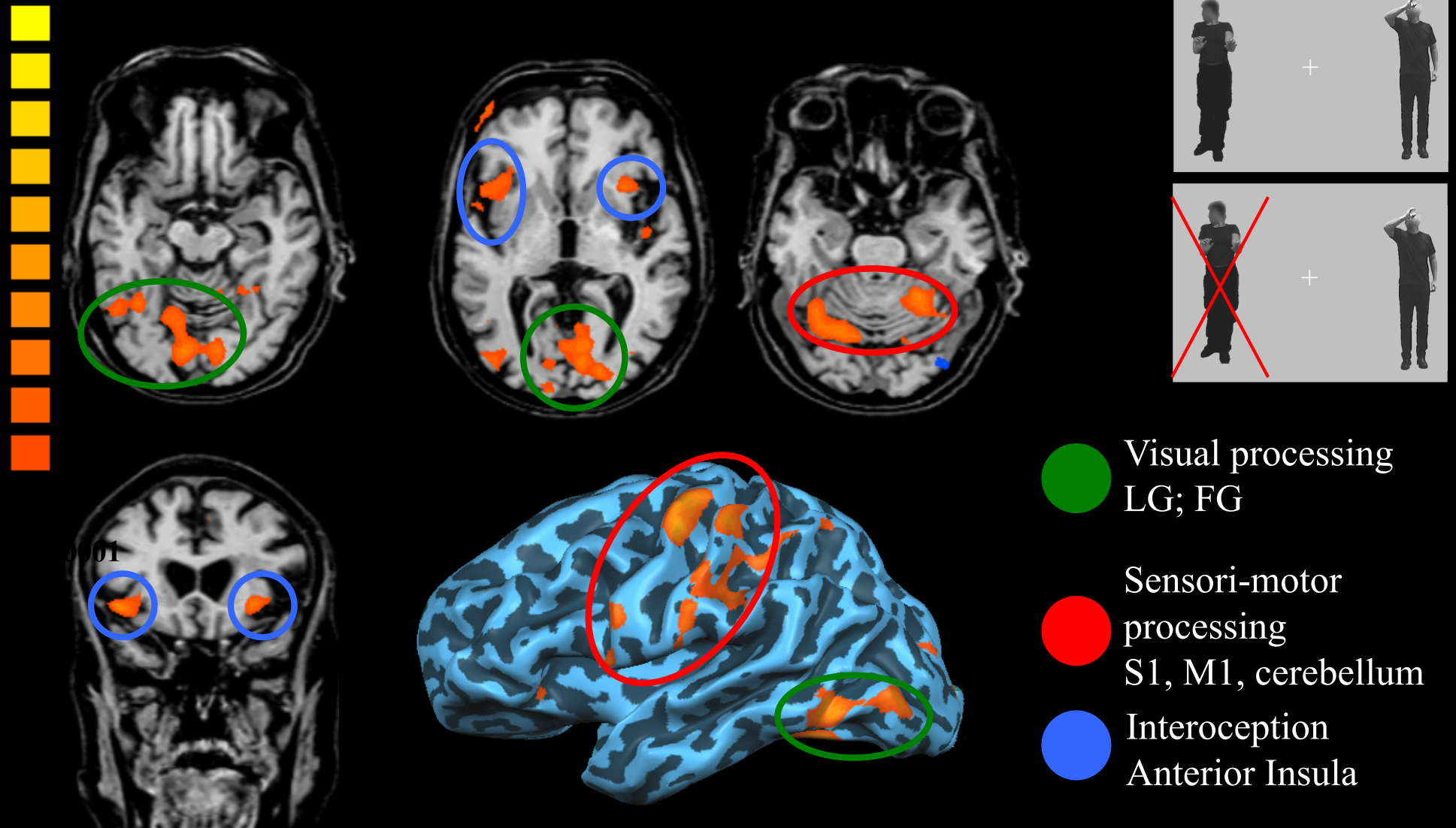
Non conscious processing of fearful bodies



- Body Process
MTG; EBA
- Emotional Process
AMG; OFC, PCC
- Attentional shift, and
action planning
TPJ (PPC); MFG

Tamietto et al. (2015). *Cortex*

Conscious processing of fearful bodies



Attentional Unawareness for emotions in neglect

. What is the neural fate of unattended emotions and the neural fingerprint of attentional awareness for emotions?

Neglected fearful bodied still activate a panel of brain areas consistent with the nature of the stimulus

- The central neural mapping of peripheral visceral and sensory-motor changes in the insula, somatosensory and motor cortices is uniquely associated to visual awareness for emotion under attentional comptetion***